Introduction to Programming:

OVERVIEW OF C PROGRAMMING LANGUAGE

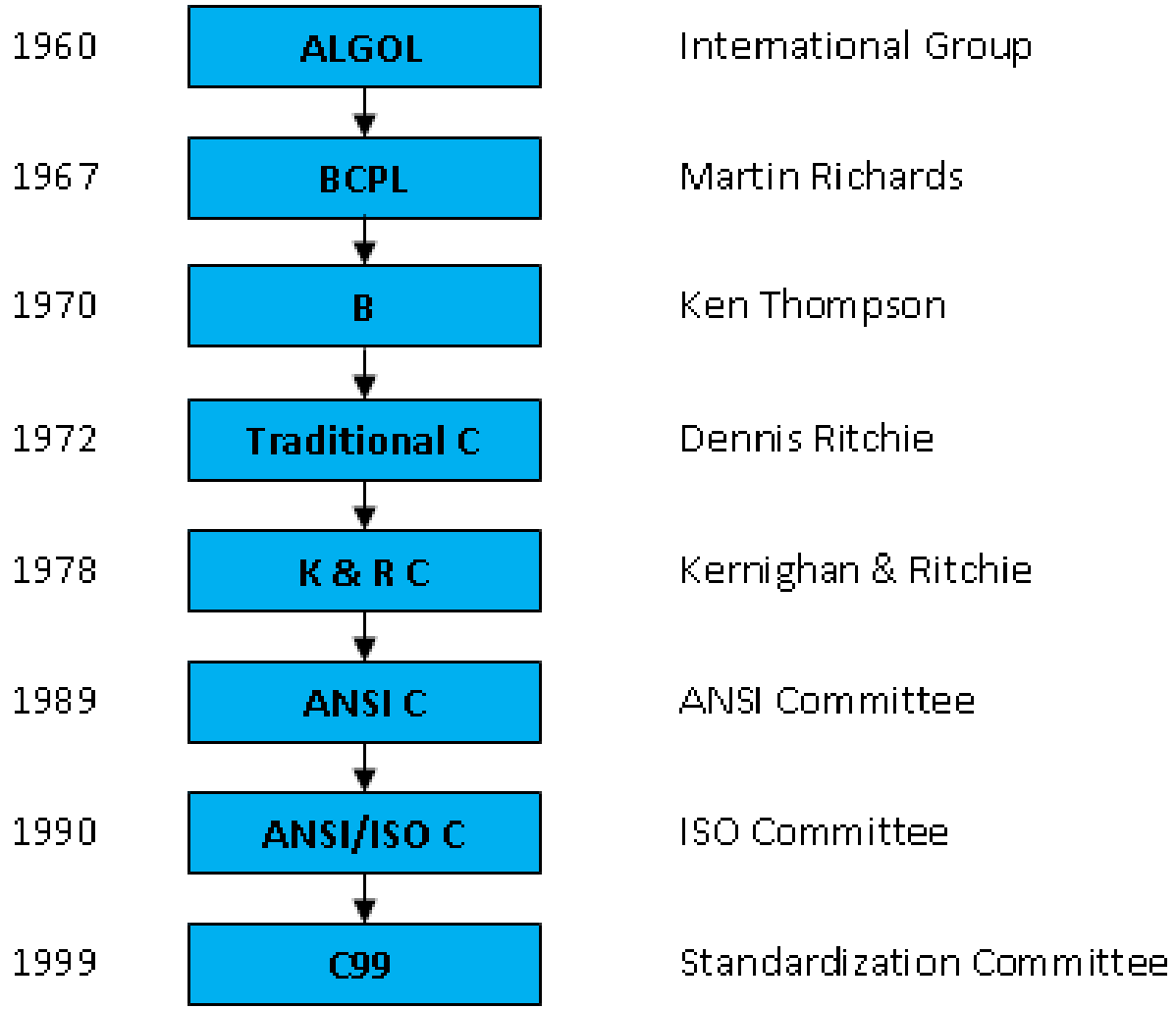
Prepared By:

Darsha Chauhan

Outline

- History of C
- Features of C
- Flowchart
- Algorithm
- Basic Structure of a C Program
- Executing a C Program

History of C


1960
ALGOL
International Group
1967
BCPL
Martin Richards
1970
B
Ken Thompson
1972
Traditional C
Dennis Ritchie
1978
K & R C
Kernighan & Ritchie
1989
ANSI C
ANSI Committee
1990
ANSI/ISO C
ISO Committee
1999
C99
Standardization Committee

Features of C

- Modularity
- Extensibility
- Elegant syntax
- Case sensitive
- Less memory required
- The standard library concept
- The portability of the compiler
- A powerful and varied range of operators
- Ready access to the hardware when needed



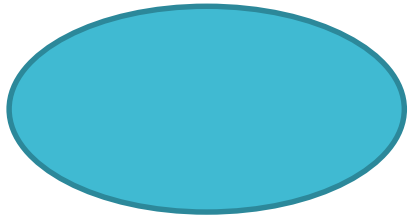
Flowchart

- Flowchart is a pictorial or graphical representation of a program.
- It is drawn using various symbols.
- Easy to understand.
- Easy to show branching and looping.
- Flowchart for big problem is impractical.



Flowchart

Start/Stop



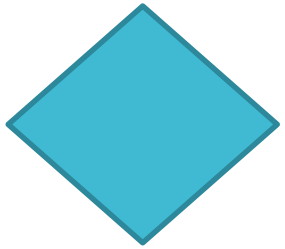
Process



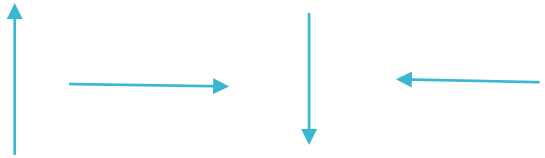
Input/Output:



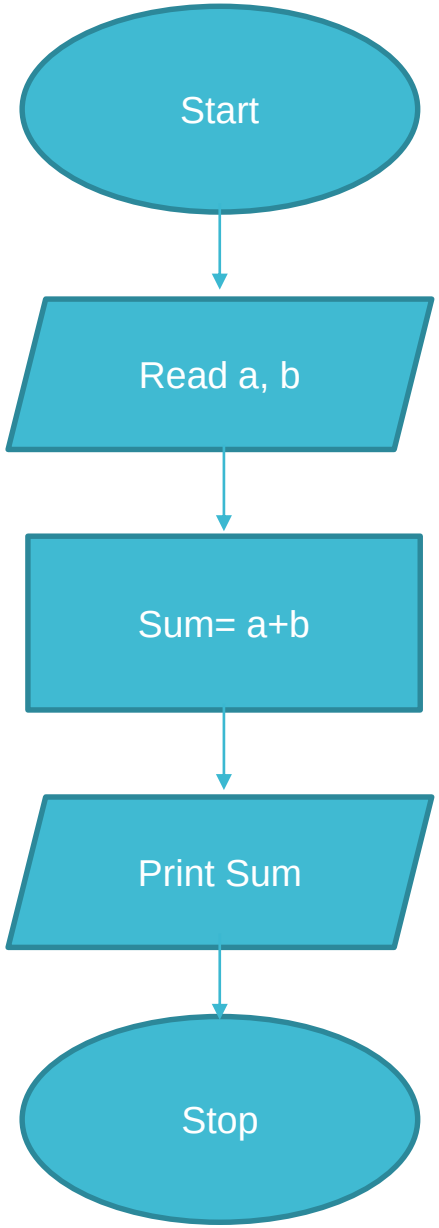
Decision



Arrow



Flowchart



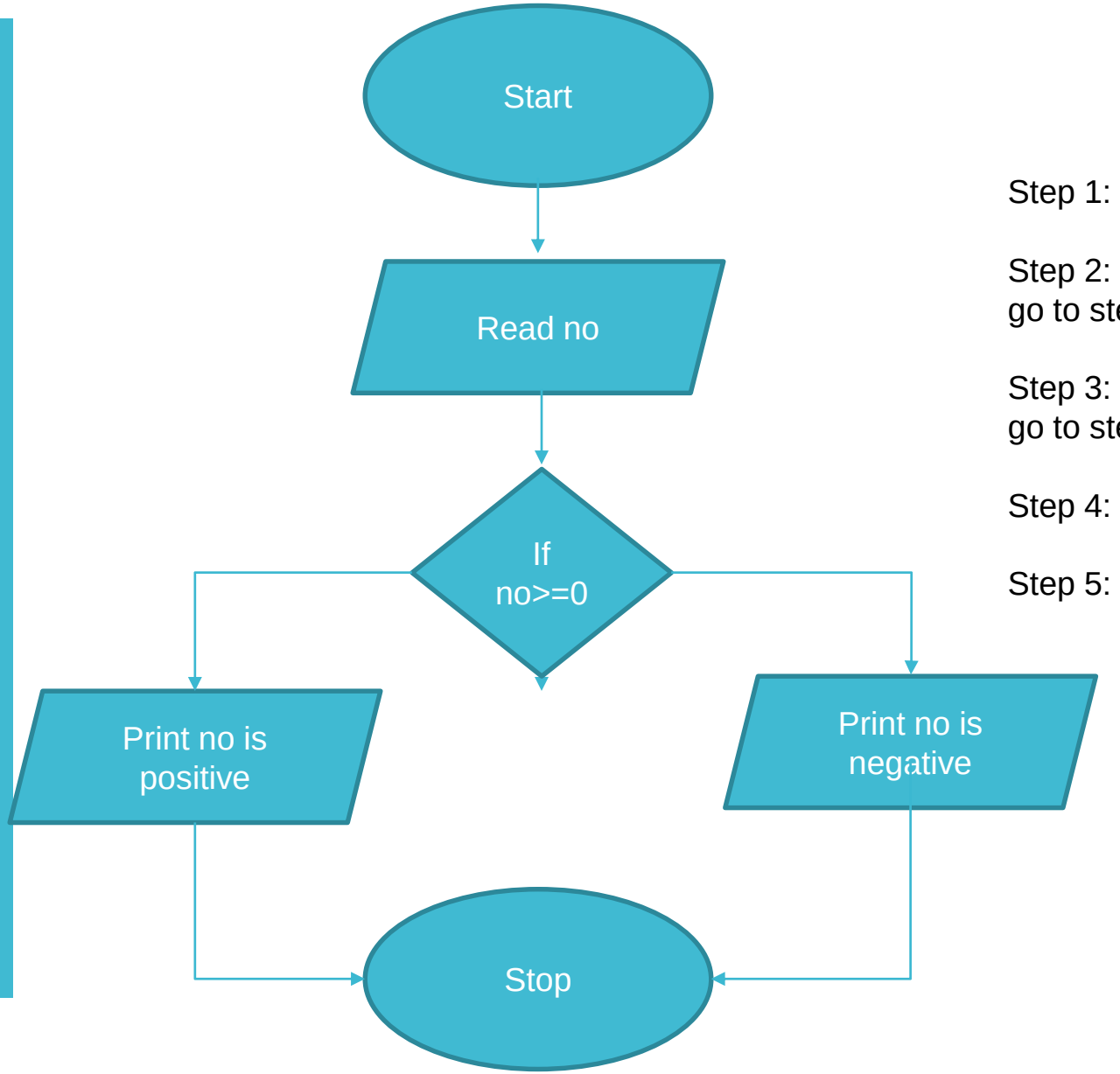
Step 1: Read a, b.

Step 2: Sum= a+b

Step 3: Print Sum

Step 4: Stop.

Flowchart



Step 1: Read no.

Step 2: If no is greater than equal zero, go to step 4.

Step 3: Print no is a negative number, go to step 5.

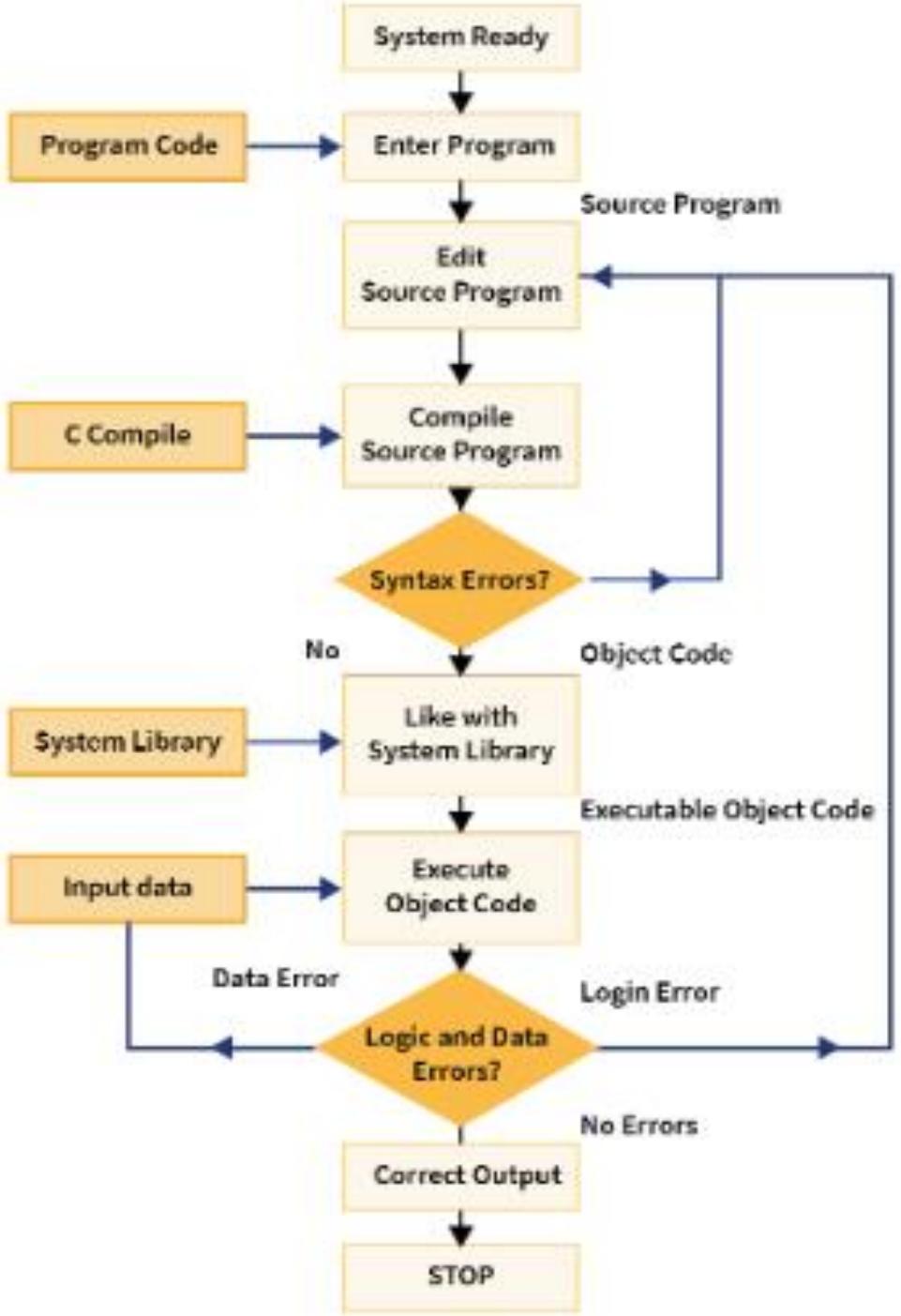
Step 4: Print no is a positive number.

Step 5: Stop.

Algorithm

- Algorithm is a finite sequence of well defined steps for solving a problem.
- It is written in the natural language like English.
- Difficult to understand.
- Difficult to show branching and looping.
- Algorithm can be written for any problem.

Executing a C program



Basic Structure of a C program

```
Documentation section
        (Used for comments)
Link section

Definition section

Global declaration section
(Variables used in more
than  one functions)
void main ()
{
        Declaration
        part
        Executable part
}
Subprogram section
        (User defined functions)
```

```c
// Program for addition of 2 nos




void fun();

int a=10;


void main( )
{
  printf("Value of a inside main function: %d",
  a);
  fun();
}

void fun()
{printf("Value of a inside fun function: %d", a);}
```

Basic Structure of a C program

The Documentation Section:

- It consist a set of comment lines giving the name of the program, the author and other details, which the programmer would like to use later.

The Link Section:

- It provides instructions to the Compiler to link functions from the system library.

The Definition Section:

- It defines all symbolic constant.

Global Declaration Section:

- There are some variables that are used in more than one function.
- Such variables are called global variables and are declared in the global declaration section that is outside of all functions.
- This section also declares all the user-defined functions.



Basic Structure of a C program

Main() Function Section:

- Every C program must have one main () function section.
- This section contains two parts, **declaration part** and **executable part**.
- **The declaration part:** It declares all the variables used in executable part.
- **Executable part**: There is at least one statement in executable part.
- These two parts must appear between the opening and closing braces.
- The program execution begins at the opening brace and ends at the closing brace.
- The closing brace of a main function is the logical end of a program. All the statements in the declaration and executable parts end with semicolon (;).

The Subprogram Section:

- It contains all the user-defined functions that are called in the main function. The user-defined function are generally placed immediately after the main function, although the may appear in any order.
- All sections, except the main function section may be absent when they are not required.



printf()

- It is a predefined standard C function for printing output
- The printf() causes everything between the starting and the ending quotation mark to be printed out.
- Syntax:

```
printf(" To print ");
```

- Example:

```
printf(" Hello World");
printf(" ");
 printf("%d", c);                    // 15
printf("Sum = %d", c);               // Sum = 15
printf("%d+%d=%d", a, b, c);         // 10+5=15
```



scanf()

- It is a predefined standard C function for taking input
- Syntax:

  scanf(" Control String ", &variable1, &variable2,...);

- Example:

  int a;

  scanf("%d",&a);



Comments

- A comment is an explanation or description of the source code of the program
- It helps a programmer to explain logic of the code and improves program readability.
- At run-time, a comment is ignored by the compiler.
- There are two types of comments in C:
  - Single line comment:
    - Represented as // double forward slash
    - It is used to denote a single line comment only.
    - Example: // Single line comment
  - Multi-line comment:
    - Represented as /* any_text */ start with forward slash and asterisk (/*) and end with asterisk and forward slash (*/).
    - It is used to denote single as well as multi-line comment.
    - Example: /* multi line comment line -1
      multi line comment line -2 */



Header Files

- In C programming, a source code file that contains common functions and types that all programmer may use as required is called header files.
- Header files is where these standardized and commonly used functions are defined.
- A header file is a file with extension .h which contains the set of predefined standard library functions.
- The "#include" preprocessing directive is used to include the header files with extension in the program.

Header File	Description
stdio.h	Standard Input/Output functions (printf() and scanf())
conio.h	Console Input/Output functions (getch() and clrscr())
math.h	Mathematics functions (pow(), exp(), sqrt() etc…)
string.h	String functions (strlen(), strcmp(), strcat() etc…)

#Thank You

From @:Darsha

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